

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA



$$\begin{cases} P_{max} = \frac{E_{TH}^2}{4R_{TH}} \\ R_N = R_{eq} = R_{TH} \end{cases}$$

$$V_{AB} = \frac{R_N}{R_{TH} + R_N} \cdot E_{TH}$$

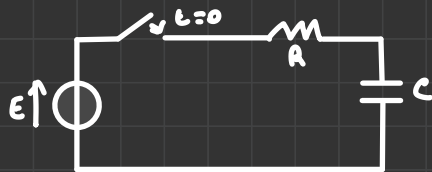
$$i = \frac{V_{AB}}{R_N} = \frac{E_{TH}}{R_{TH} + R_N}$$

$$P = R_N \cdot i^2 = R_N \cdot \frac{E_{TH}^2}{(R_{TH} + R_N)^2} = R_N \cdot \frac{E_{TH}^2}{(R_{TH}^2 + R_N^2 + 2R_{TH}R_N)} = \frac{E_{TH}^2}{\frac{R_{TH}^2}{R_N} + R_N + 2R_{TH}}$$

$$\frac{d}{dR_N} \left(\frac{R_{TH}^2}{R_N} + R_N + 2R_{TH} \right) = -\frac{R_{TH}^2}{R_N^2} + 1 + 0 \rightarrow R_{TH} = R_N$$

$$P_{max} = R_{TH} \cdot \frac{E_{TH}^2}{(2R_{TH})^2} = \frac{E_{TH}^2}{4R_{TH}}$$

CARICA E SCARICA DEL CONDENSATORE



$$i(t) = \begin{cases} 0 & t \leq 0^- \\ i(t) & t > 0^+ \end{cases}$$

$$e(t) = E u_{-1}(t)$$

$$e(t) = u_R(t) + u_C(t) \quad t \geq 0^+$$

$$E = R \cdot i + u_C = RC \frac{du_C}{dt} + u_C$$

$$u_C(t) = u_{C_f}(t) + u_{C_p}(t)$$

$$RC \frac{du_C}{dt} + u_C = 0 \rightarrow RC \alpha + 1 = 0$$

$$\alpha = -\frac{1}{RC} \quad u_{C_f}(t) = k_1 e^{-\frac{1}{RC}t}$$

$$u_{C_p}(t) = k_2 \quad u_C(t) = k_1 e^{-\frac{1}{RC}t} + k_2$$

$$\begin{cases} u_C(0^-) = k_1 + k_2 \\ \lim_{t \rightarrow \infty} u_C(t) = E = k_2 \end{cases}$$

$$\begin{cases} k_1 = u_C(0^-) - E \\ k_2 = E \end{cases}$$

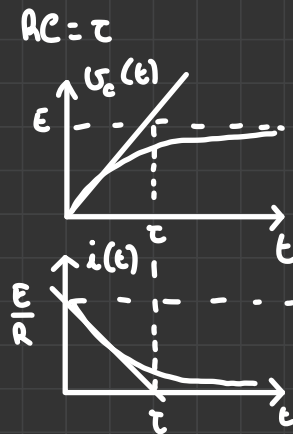
$$\begin{cases} u_C(t) = (u_C(0^-) - E) e^{-\frac{1}{RC}t} + E \\ i(t) = -\frac{1}{R} (u_C(0^-) - E) e^{-\frac{1}{RC}t} \end{cases}$$

CARICA $|u_C(0^-)| < |E|$

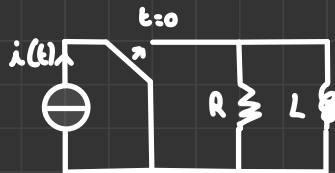
SCARICA $|u_C(0^-)| > |E|$

PER $u_C(0^-) = 0$

$$\begin{aligned} u(t) &= E (1 - e^{-\frac{1}{RC}t}) u_{-1}(t) \\ i(t) &= \frac{E}{R} e^{-\frac{1}{RC}t} \end{aligned}$$



CARICA E SCARICA DELL'INDUTTORE



$$i(t) - G U(t) - i_L(t) = 0$$

$$i(t) = I u_{-1}(t) \quad t \geq 0^+$$

$$I = GL \frac{di_L}{dt} + i_L$$

$$i_L(t) = i_{Lg}(t) + i_{Lp}(t)$$

$$GL \frac{di_L}{dt} + i_L = 0 \rightarrow GL \alpha + 1 = 0$$

$$\alpha = -\frac{1}{GL} \quad i_{Lg}(t) = k_1 e^{-\frac{1}{GL}t}$$

$$i_{Lp}(t) = k_2 \quad i_L(t) = k_1 e^{-\frac{1}{GL}t} + k_2$$

$$\begin{cases} i_L(0^+) = k_1 + k_2 \\ \lim_{t \rightarrow \infty} i_L(t) = I = k_2 \end{cases}$$

$$\begin{cases} k_1 = i_L(0^+) - I \\ k_2 = I \end{cases}$$

$$\begin{cases} i_L(t) = (i_L(0^+) - I) e^{-\frac{1}{GL}t} + I \\ U(t) = -\frac{1}{G} (i_L(0^+) - I) e^{-\frac{1}{GL}t} \end{cases}$$

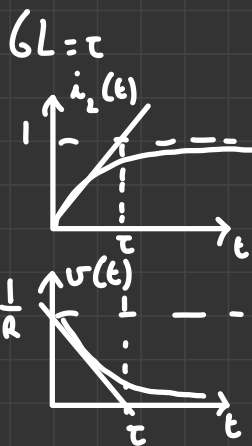
CARICA $|i_L(0^+)| < I$

SCARICA $|i_L(0^+)| > I$

$$i_L(0^+) = 0$$

$$i_L(t) = I (1 - e^{-\frac{1}{GL}t}) u_{-1}(t)$$

$$U(t) = \frac{1}{G} e^{-\frac{1}{GL}t}$$



CIRCUITI IN REGIME SINUSOIDALE E FASORI

$$v(t) = V_m \sin(\omega t + \alpha) = \frac{V_m}{2j} (e^{j(\omega t + \alpha)} - e^{-j(\omega t + \alpha)})$$

$$i(t) = I_m \sin(\omega t + \beta) = \frac{I_m}{2j} (e^{j(\omega t + \beta)} - e^{-j(\omega t + \beta)})$$

FASORI

$$\bar{V} = \frac{V_m e^{j\alpha}}{2j} \quad \bar{I} = \frac{I_m e^{j\beta}}{2j}$$

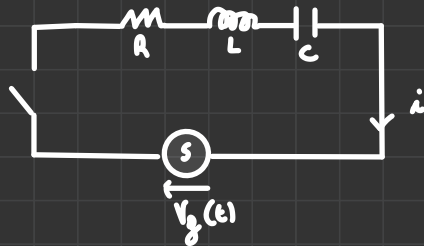
$$v(t) = \frac{V_m}{2j} (e^{j(\omega t + \alpha)} - e^{-j(\omega t + \alpha)}) = \bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}$$

$$i(t) = \frac{I_m}{2j} (e^{j(\omega t + \beta)} - e^{-j(\omega t + \beta)}) = \bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}$$

$$\text{PER } \bar{V} = V_m e^{j\alpha} \quad \bar{I} = I_m e^{j\beta}$$

$$v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j} \quad i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

FASORI NEI CIRCUITI RLC CON FORZANTE SINUSOIDALE



$$V_g(t) = V_M \sin(\omega t + \sigma)$$

$$\begin{cases} V_g(t) = R i(t) + L \frac{di(t)}{dt} + v_C(t) \\ i(t) = C \frac{dv_C(t)}{dt} \end{cases}$$

$$v_{gp}(t) = V_M \sin(\omega t + \theta) \quad v_{gp}(0) = V_M \sin(\theta)$$

$$i(t) = I_M \sin(\omega t + \beta)$$

$$v_{gp}(t) = \frac{1}{C} \int_0^t i(t) dt + v_{C(0-)} \quad \text{or} \quad v_{C(0-)} = 0$$

$$v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j} \quad i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

$$\bar{V}_g = R \bar{I} + j\omega L \bar{I} + \frac{1}{j\omega C} \bar{I} = (R + j(\omega L - \frac{1}{\omega C})) \bar{I}$$

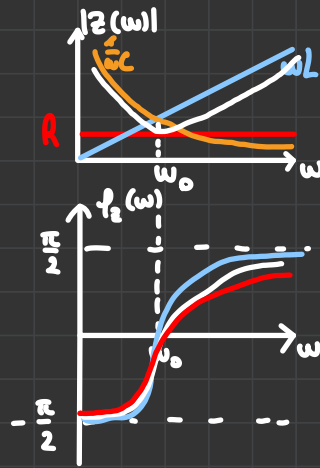
$$z = R + j(\omega L - \frac{1}{\omega C}) \quad \bar{V} = z \bar{I}$$

STUDIAMO L'IMPIEDENZA AL VARIARE DI ω NEI CIRCUITI RLC SERIE

$$Z(\omega) = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$|Z(\omega)| = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}$$

$$\varphi_2(\omega) = \text{Arctan}\left(\frac{\omega L - \frac{1}{\omega C}}{R}\right)$$



$\omega > \omega_0$ OHMICO CAPACITIVO

SE $R=0$ CAPACITIVO

$\omega < \omega_0$ OHMICO-INDUTTIVO

SE $R=0$ INDUTTIVO

$\omega \rightarrow \infty$ $\omega L \gg \frac{1}{\omega C}$

$\omega = \omega_0$ $\varphi_2 = 0$

$\omega \rightarrow 0$ $\frac{1}{\omega C} \gg \omega L$

POTENZA NEL REGIME SINUSOIDALE

$$v(t) = V_M \sin(\omega t + \sigma) \longrightarrow v(t) = \frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j}$$

$$i(t) = I_M \sin(\omega t + \beta) \longrightarrow i(t) = \frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j}$$

$$p(t) = v(t)i(t) = \left(\frac{\bar{V} e^{j\omega t} - \bar{V}^* e^{-j\omega t}}{2j} \right) \left(\frac{\bar{I} e^{j\omega t} - \bar{I}^* e^{-j\omega t}}{2j} \right) = -\frac{1}{4} (\bar{V} \bar{I} e^{2j\omega t} - \bar{V} \bar{I}^* - \bar{V}^* \bar{I} + \bar{V}^* \bar{I} e^{-2j\omega t})$$

$$= -\frac{1}{2} \frac{\bar{V} \bar{I} e^{2j\omega t} + \bar{V}^* \bar{I}^* e^{-2j\omega t}}{2} + \frac{1}{2} \frac{\bar{V} \bar{I}^* + \bar{V}^* \bar{I}}{2}$$

VAR. NEL TEMPO COSTANTE

$$\bar{V} \bar{I} = V_M I_M e^{j(\sigma + \beta)} \quad \bar{V} \bar{I}^* = V_M I_M e^{j(\sigma - \beta)}$$

$$(\bar{V} \bar{I})^* = V_M I_M e^{-j(\sigma + \beta)} \quad \bar{V}^* \bar{I} = V_M I_M e^{-j(\sigma - \beta)}$$

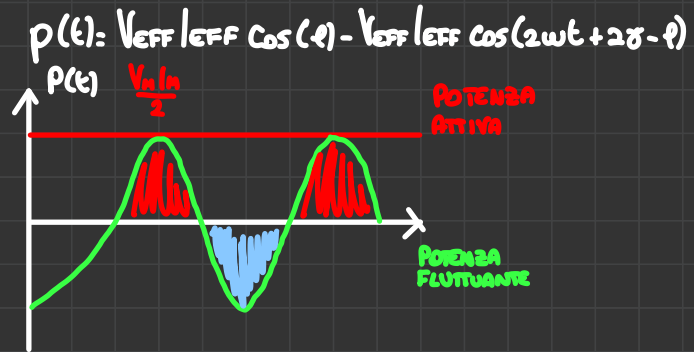
$$\sigma - \beta = \varphi$$

$$\sigma + \beta = 2\sigma + \varphi$$

$$p(t) = -\frac{1}{2} V_M I_M \frac{e^{j(2\sigma t + 2\sigma + \varphi)} + e^{-j(2\sigma t + 2\sigma + \varphi)}}{2} + \frac{1}{2} V_M I_M \frac{e^{j\varphi} - e^{-j\varphi}}{2}$$

$$= -\frac{1}{2} V_M I_M \cos(2\omega t + 2\sigma + \varphi) + \frac{1}{2} V_M I_M \cos(\varphi)$$

$$\begin{cases} V_{\text{eff}} = \frac{V_M}{\sqrt{2}} \longrightarrow V_M = \sqrt{2} V_{\text{eff}} \\ I_{\text{eff}} = \frac{I_M}{\sqrt{2}} \longrightarrow I_M = \sqrt{2} I_{\text{eff}} \end{cases}$$



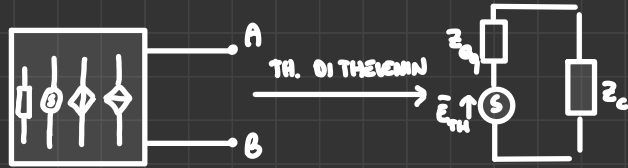
$$P_c = \bar{V} \bar{I}^*$$

$$P_c = P + jQ$$

$$P = |P_c| \cos(\varphi)$$

$$Q = |P_c| \sin(\varphi)$$

TEOREMA DEL MASSIMO TRASFERIMENTO DI POTENZA PER RETI DINAMICHE



$$z_c = z_{eq}^*$$

$$\bar{I} = \frac{\bar{E}_{TH}}{z_{eq} + z_c} = \frac{\bar{E}_{TH}}{(R_{eq} + R_c) + j(X_{eq} + X_c)}$$

$$P_c: z \bar{I}^2 = R \bar{I}^2 + jX \bar{I}^2$$

$$P_c = R_c \bar{I}^2 = R_c \frac{\bar{E}_{TH}^2}{(R_{eq} + R_c)^2 + (X_{eq} + X_c)^2}$$

$$X_c = -X_{eq} \quad X_{tot} = 0$$

FILTRO PASSA-BASSO PASSIVO DEL II ORDINE

$$H(s) = \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{1}{sCR + s^2LC + 1} = \frac{1}{LC(s\frac{R}{L} + s^2 + \frac{1}{LC})} = \frac{\omega_0^2}{(Bs + s^2 + \omega_0^2)}$$

$$\omega_0 = \sqrt{\frac{1}{LC}} \quad B = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{\omega_0^2}{(Bj\omega - \omega^2 + \omega_0^2)}$$

$$|H(j\omega)| = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (B\omega)^2}}$$

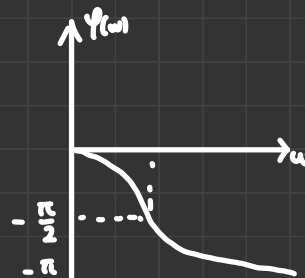
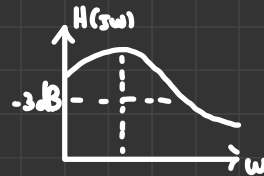
$$\varphi(\omega) = \begin{cases} \arctan\left(-\frac{\omega_0^2 - \omega^2}{B\omega}\right) + \pi & \omega \leq \omega_0 \\ \arctan\left(-\frac{\omega_0^2 - \omega^2}{B\omega}\right) & \omega > \omega_0 \end{cases}$$

$$\varphi(0) = 0$$

$$\varphi(\omega_0) = -\frac{\pi}{2}$$

$$\varphi(\infty) = -\pi$$

$$\begin{aligned} \omega &\rightarrow 0 & H(j\omega) &\rightarrow 1 \\ \omega &= \omega_0 & H(j\omega) & \\ \omega &\rightarrow \infty & H(j\omega) &\rightarrow 0 \end{aligned}$$



FILTRO ELIMINA-BANDA PASSIVO DO 1º ORDEM

$$H(s) = \frac{sL + \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{s^2 LC + 1}{s^2 LC + sR + 1} = \frac{LC(s^2 + \frac{1}{LC})}{LC(s\frac{R}{L} + s^2 + \frac{1}{LC})} = \frac{s^2 + \omega_0^2}{Bs + s^2 + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{\omega_0^2 - \omega^2}{Bs + \omega_0^2 - \omega^2}$$

$$|H(j\omega)| = \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + (B\omega)^2}}$$

$$\omega \rightarrow 0 \quad H(j\omega) \rightarrow 1$$

$$\omega = \omega_0 \quad H(j\omega) \rightarrow 0$$

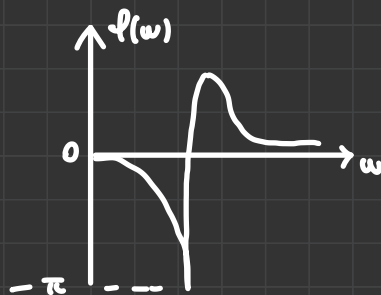
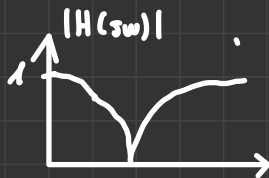
$$\omega \rightarrow \infty \quad H(j\omega) \rightarrow 1$$

$$\varphi(\omega) = \text{ARCTAN} \left(-\frac{B\omega}{\omega_0^2 - \omega^2} \right)$$

$$\varphi(0) = 0$$

$$\varphi(\omega_0) = -\frac{\pi}{2}$$

$$\varphi(\infty) = -\pi$$



FILTRO PASSA-BASSO PASSIVO DE 1º ORDEM

$$H(s) = \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{1}{sLR + s^2LC + 1} = \frac{1}{LC(s\frac{R}{L} + s^2 + \frac{1}{LC})} = \frac{\omega_0^2}{(Bs + s^2 + \omega_0^2)}$$

$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$

S = jw

$$H(jw) = \frac{\omega_0^2}{(Bjw + \omega_0^2 - w^2)}$$

MÓDULO

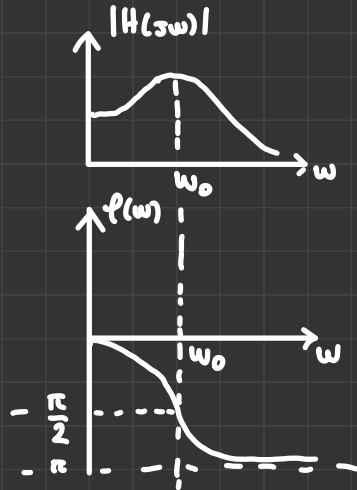
$$|H(jw)| = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - w^2)^2 + (wB)^2}}$$

FASE

$$\phi(w) = \begin{cases} \text{ARCTAN}\left(-\frac{wB}{\omega_0^2 - w^2}\right) & w \leq \omega_0 \\ \text{ARCTAN}\left(-\frac{wB}{\omega_0^2 - w^2}\right) - \pi & w > \omega_0 \end{cases}$$

$w \rightarrow 0 \quad H(jw) \rightarrow 1$
 $w = \omega_0 \quad H(jw)$
 $w \rightarrow \infty \quad H(jw) \rightarrow 0$

$\phi(0) = 0$
 $\phi(\omega_0) = -\frac{\pi}{2}$
 $\phi(\infty) = -\pi$



FILTRO PASSA-ALTO PASSIVO II ORDINE

$$H(s) = \frac{sL}{R + sL + \frac{1}{sC}} = \frac{s^2 LC}{sCR + s^2 LC + 1} = \frac{s^2 LC}{LC(s\frac{R}{L} + s^2 + \frac{1}{LC})} = \frac{s^2}{(bs + s^2 + \omega_0^2)}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{-\omega^2}{(b j\omega + \omega_0^2 - \omega^2)}$$

MODULO

$$|H(j\omega)| = \frac{\omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (b\omega)^2}}$$

FASE

$$\varphi(\omega) = \begin{cases} \text{ARCTAN}\left(-\frac{b\omega}{\omega_0^2 - \omega^2}\right) + \pi & \omega \leq \omega_0 \\ \text{ARCTAN}\left(-\frac{b\omega}{\omega_0^2 - \omega^2}\right) & \omega > \omega_0 \end{cases}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 0$$

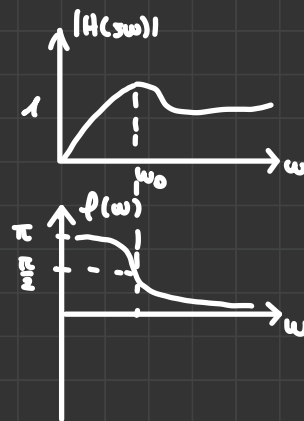
$$\omega = \omega_0 \quad |H(j\omega)|$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 1$$

$$\varphi(0) = \pi$$

$$\varphi(\omega_0) = -\frac{\pi}{2} + \pi = \frac{\pi}{2}$$

$$\varphi(\infty) = 0$$



FILTRO ELIMINA BANDA PASIVO DEL II ORDINE

$$H(s) = \frac{sL + \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{s^2 LC + 1}{s(R + s^2 LC + 1)} = \frac{\cancel{L}(s^2 + \frac{1}{LC})}{\cancel{L}(s\frac{R}{L} + s^2 + \frac{1}{LC})} = \frac{s^2 + \omega_0^2}{Bs + s^2 + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{\omega_0^2 - \omega^2}{Bs + \omega_0^2 - \omega^2}$$

MODULO

$$|H(j\omega)| = \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + (B\omega)^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 1$$

$$\omega = \omega_0 \quad |H(j\omega)| \rightarrow 0$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 1$$

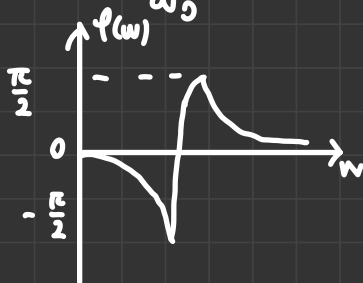
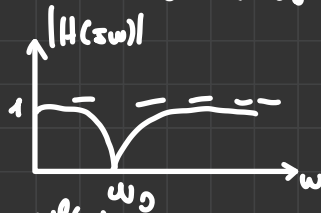
$$\phi(0) = 0$$

$$\phi(\omega_0) = -\frac{\pi}{2}$$

$$\phi(\infty) = 0$$

FASE

$$\phi(\omega) = \text{Arctan} \left(-\frac{B\omega}{\omega_0^2 - \omega^2} \right)$$



FILTRI

FILTRO PASSO BASSO PASSIVO DEL 1° ORDINE

$$H(s) = \frac{\frac{1}{sC}}{R + \frac{1}{sC}} = \frac{1}{sRC + 1} = \frac{1}{RC(s + \frac{1}{RC})}$$

$$s = j\omega$$

$$H(j\omega) = \frac{1}{RC(j\omega + \frac{1}{RC})}$$

MODULO

$$|H(j\omega)| = \frac{1}{RC \sqrt{\omega^2 + (\frac{1}{RC})^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 1$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 0$$

FASE

$$\varphi(\omega) = \text{ARCTAN}(-\omega RC)$$

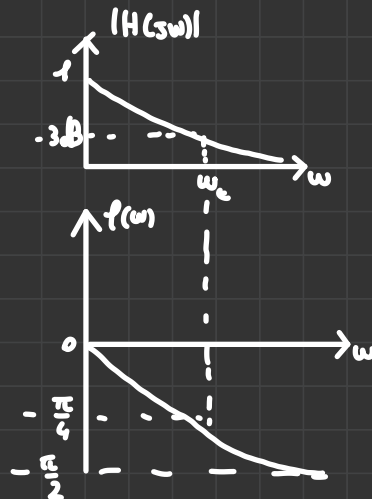
OMEGA DI TAGLIO

$$|H(j\omega)| = \frac{1}{\sqrt{2}} = -3 \text{ dB}$$

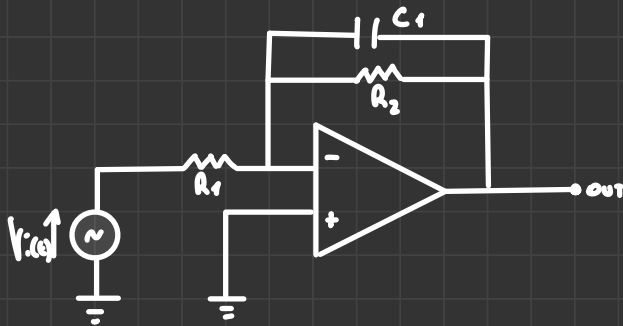
$$\frac{1}{RC \sqrt{\omega_c^2 + (\frac{1}{RC})^2}} = \frac{1}{\sqrt{2}} \rightarrow \omega_c^2 + (\frac{1}{RC})^2 = 2(\frac{1}{RC})^2 \rightarrow \omega_c = \frac{1}{RC}$$

$$H(j\omega) = \frac{\omega_c}{s + \omega_c}$$

$$\varphi(\omega_c) = \text{ARCTAN}\left(-\frac{1}{RC} RC\right) = \text{ARCTAN}(-1) = -\frac{\pi}{4}$$



FILTRO PASSA-BASSO ATTIVO DEL 1° ORDINE



$$H(s) = - \frac{G_1}{sC_1 + G_2} = - \frac{\frac{G_1}{C_1}}{s + \frac{G_2}{C_1}} = - \frac{\omega_c}{s + \omega_c}$$

$$\text{con } \omega_c = \frac{G_2}{C_1} \quad \text{e } G_1 = G_2$$

FILTRO PASSA-ALTO PASSIVO DEL 1° ORDINE

$$H(s) = 1 - H_{LP}(s) = 1 - \frac{\omega_c}{s + \omega_c} = \frac{s}{s + \omega_c}$$

$$\omega_c = \frac{1}{RC}$$

$$s = j\omega$$

$$H(j\omega) = \frac{j\omega}{j\omega + \omega_c}$$

MODULO

$$|H(j\omega)| = \frac{\omega}{\sqrt{(\omega)^2 + (\omega_c)^2}}$$

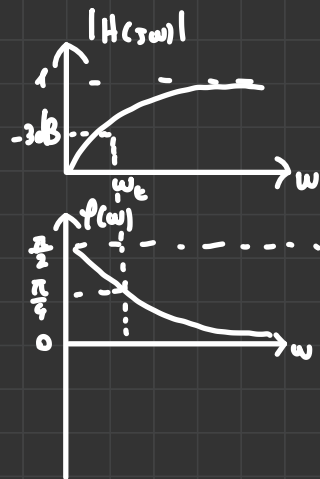
$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 0$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 1$$

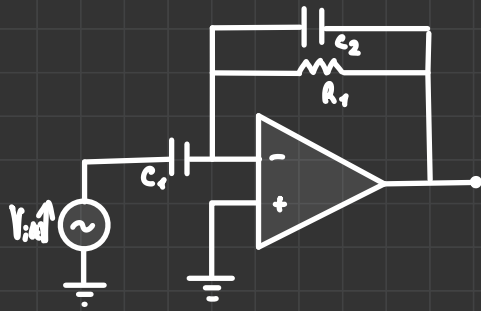
FASE

$$\varphi(\omega) = \text{ARCTAN} \left(\frac{\omega}{\omega_c} \right)$$

$$\varphi(\omega_c) = \text{ARCTAN}(1) = \frac{\pi}{4}$$



FILTRO PASSA-ALTO ATTIVO DEL 1° ORDINE



$$H(s) = -\frac{C_1}{C_2} \frac{s}{s + \frac{G_1}{C_2}} = -\frac{s}{s + \omega_c}$$

$$\omega_c = \frac{G_1}{C_2} \quad C_1 = C_2$$

FILTRO PASSA BANDA PASSIVO DEL II ORDINE

$$H(s) = \frac{R}{R + sL + \frac{1}{sC}} = \frac{sCR}{sCR + s^2LC + 1} = \frac{sCR}{L(s^2 + s^2 + \frac{1}{LC})} \quad \omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L} = \frac{B_s}{(B_s + s^2 + \omega_0^2)}$$

$$s = j\omega$$

$$H(j\omega) = \frac{B_s \omega}{B_s \omega + \omega_0^2 - \omega^2}$$

MODULO

$$|H(j\omega)| = \frac{\omega B_s}{\sqrt{(\omega_0^2 - \omega^2)^2 + (\omega B_s)^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 0$$

$$\omega = \omega_0 \quad |H(j\omega)| \rightarrow 1$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 0$$

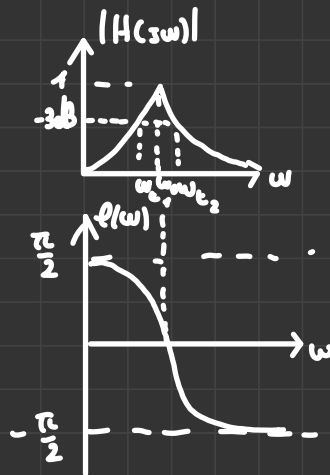
FASE

$$\varphi(\omega) = A \arctan\left(\frac{\omega_0^2 - \omega^2}{\omega B_s}\right)$$

$$\varphi(0) = \frac{\pi}{2}$$

$$\varphi(\omega_0) = 0$$

$$\varphi(\infty) = -\frac{\pi}{2}$$



FILTRO PASSA-BASSO PASSIVO DEL II ORDINE

$$H(s) = \frac{\frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{1}{sCRs^2LC + 1} = \frac{1}{LC(s^2 + \frac{R}{L}s + \frac{1}{LC})} = \frac{\omega_0^2}{Bs^2 + s^2 + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{\omega_0^2}{Bs^2 + \omega_0^2 - \omega^2}$$

MODULO

$$|H(j\omega)| = \frac{\omega_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (B\omega)^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 1$$

$$\omega = \omega_0 \quad |H(j\omega)|$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 0$$

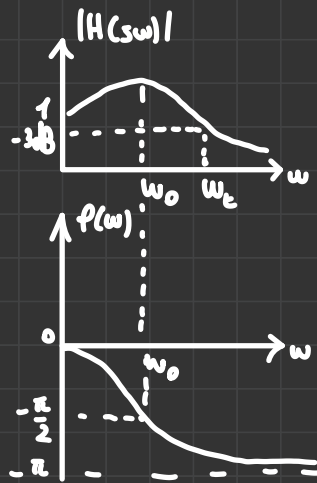
FASE

$$\varphi(\omega) = \begin{cases} \text{ARCTAN} \left(-\frac{B\omega}{\omega_0^2 - \omega^2} \right) & \omega \leq \omega_0 \\ \text{ARCTAN} \left(-\frac{B\omega}{\omega_0^2 - \omega^2} \right) - \pi & \omega > \omega_0 \end{cases}$$

$$\varphi(0) = 0$$

$$\varphi(\omega_0) = -\frac{\pi}{2}$$

$$\varphi(\infty) = -\pi$$



FILTRO PASSA-ALTO PASSIVO DEL II ORDINE

$$H(s) = \frac{sL}{R + sL + \frac{1}{sC}} = \frac{s^2 LC}{s^2 LC + 1} = \frac{s^2 LC}{LC(s^2 R + s^2 + \frac{1}{LC})} = \frac{s^2}{Bs + s^2 + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{-\omega^2}{Bs + \omega_0^2 - \omega^2}$$

MODULO

$$|H(j\omega)| = \frac{\omega^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + (B\omega)^2}}$$

$$\omega \rightarrow 0 \quad |H(j\omega)| \rightarrow 0$$

$$\omega = \omega_0 \quad |H(j\omega)|$$

$$\omega \rightarrow \infty \quad |H(j\omega)| \rightarrow 1$$

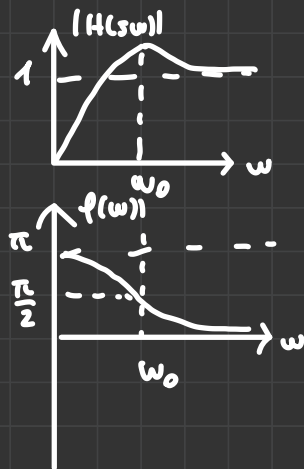
FASE

$$\varphi(\omega) = \begin{cases} \arctan\left(-\frac{B\omega}{\omega_0^2 - \omega^2}\right) + \pi & \omega < \omega_0 \\ \arctan\left(-\frac{B\omega}{\omega_0^2 - \omega^2}\right) & \omega > \omega_0 \end{cases}$$

$$\varphi(0) = \pi$$

$$\varphi(\omega_0) = \frac{\pi}{2}$$

$$\varphi(\infty) = 0$$



FILTRO ELIMINA-BANDA PASIVO DEL 1º ORDINE

$$H(s) = \frac{sL + \frac{1}{sC}}{R + sL + \frac{1}{sC}} = \frac{s^2 LC + 1}{sCR + s^2 LC + 1} = \frac{LC(s^2 + \frac{1}{LC})}{LC(s\frac{R}{L} + s^2 + \frac{1}{LC})} = \frac{s^2 + \omega_0^2}{s + s^2 + \omega_0^2}$$

$$\omega_0 = \frac{1}{\sqrt{LC}} \quad B = \frac{R}{L}$$

$$s = j\omega$$

$$H(j\omega) = \frac{\omega_0^2 - \omega^2}{B + \omega_0^2 - \omega^2}$$

MODULO

$$|H(j\omega)| = \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + (B)^2}}$$

$$\begin{aligned} \omega \rightarrow 0 & \quad |H(j\omega)| \rightarrow 1 \\ \omega = \omega_0 & \quad |H(j\omega)| \rightarrow 0 \\ \omega \rightarrow \infty & \quad |H(j\omega)| \rightarrow 1 \end{aligned}$$

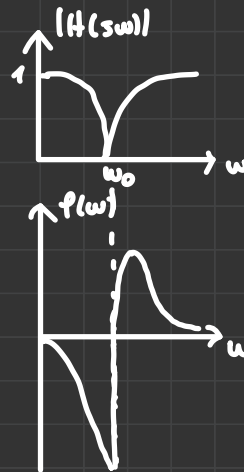
FASE

$$\varphi(\omega) = \text{ARCTAN}\left(-\frac{B}{\omega_0^2 - \omega^2}\right)$$

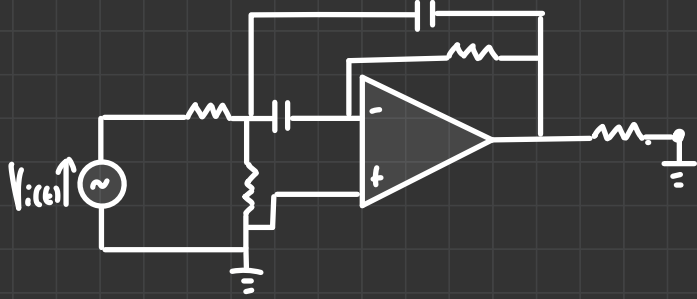
$$\varphi(0) = 0$$

$$\varphi(\omega_0) = -\frac{\pi}{2}$$

$$\varphi(\infty) = 0$$



FILTRO PASSA BANDA ATTIVO DEL II ORDINE



$$H(s) = \frac{Bs}{Bs^2 + \omega_0^2}$$